Energy Causes Change

Chapter 1 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION **Where can we observe evidence of energy causing change?**

KEY WORDS

energy — Energy is the ability to cause _____ .

change — Change means something becomes _____ .

evidence — Evidence is something we observe that helps us _____ what happened.

MODEL IT

Energy causes a change. The change is our evidence.

ENERGY → **CHANGE** → **EVIDENCE**

In my own words:

_____

DRAW / LABEL

Pick one change from class. Draw what it looked like before and after, then write what changed.

BEFORE	AFTER

What changed? _____

Forms of Energy + Stored Energy

Chapter 1 Guided Notes · Page 2

Name: _____ **Date:** _____

FORMS OF ENERGY

Energy has different forms. Write or draw one example that shows a change for each.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	
sound	
light	
heat	
electrical energy	

STORED ENERGY

KEY IDEA **Stored energy can cause a change later.**

For each example, write the change that could happen later.

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	
stretched rubber band	
raised object	

DRAW ONE STORED-ENERGY EXAMPLE

draw your stored-energy example here

Energy Detectives — Evidence

Chapter 1 Guided Notes · Page 3

Name: _____ **Date:** _____

OBSERVATION VS EXPLANATION

Observation = what I see. Explanation = what it means.

Example: The light turned on. That is an _____ because I saw the change.

Example: Energy caused the change. That is an _____ because it tells what it means.

EVIDENCE TABLE

Record one change you found as an Energy Detective.

PROMPT	YOUR ANSWER
Before	_____
After	_____
What changed?	_____
What is the evidence?	_____

CHECK YOUR THINKING

How do scientists know energy is involved if energy itself cannot be seen directly?

_____

REMEMBER

ENERGY → **CHANGE** → **EVIDENCE**

CONNECT *Chapter 1 connection: every change we observe is a clue that energy was involved.*